

Dominic Chomchai

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139 W Wilson St, Apt 205, Madison, WI, 53703

EDUCATION

University of Wisconsin-Madison, Madison, WI

August 2020 – Present

- Ph.D. candidate in the Cellular and Molecular Biology Graduate Program
- GPA: 3.864/4.0

University of Michigan, Ann Arbor, MI

September 2016 – May 2020

- Molecular, Cellular, and Developmental Biology, B.S. (Honors)
- Spanish Minor
- GPA: 3.664/4.0

RESEARCH EXPERIENCE

Ph.D. Candidate, University of Wisconsin-Madison, Madison, WI

December 2020 – Present

- Research Advisor: William M. Bement, Ph.D.
- Thesis: The basis of positive feedback during cytokinesis
- Lab Supplies: Manage the ordering of all lab supplies by communicating with department administration and company representatives.

Undergraduate Research Assistant, University of Michigan, Ann Arbor, MI

March 2018 – May 2020

- Research Advisor: Ming Li, Ph.D.
- Thesis: The mechanism of the ubiquitination and degradation of lysosomal membrane proteins
 - Independently conducted a whole genome high-throughput screening to identify proteins involved in the assembly of the vacuole membrane protein complex Fth1-Fet5 using fluorescent microscopy in *Saccharomyces cerevisiae*.
 - Studied the importance of two distinct lysosomal membrane protein degradation pathways using fluorescent microscopy, ubiquitin blot analysis, and Western blot analysis.
 - Studied the effects of different stress conditions on the general composition of lysosome membrane proteins.
 - Conducted a genetic screening of PY motif-containing proteins and analyzed with flow cytometry to identify new adaptors involved in the ubiquitination of lysosomal membrane cargoes.

OUTREACH EXPERIENCE

Special Olympics College at the University of Michigan, Ann Arbor, MI

September 2016 – May 2020

Vice President (May 2019 – May 2020)

- Collaborated with other board members to operate the organization.
- Conducted weekly board member meetings.
- Mentored new board members in their specific roles.
- Recruited intellectually and developmentally disabled athletes by contacting several other special needs organizations in Ann Arbor.
- Organized and conducted a booth at the student organization fair every fall and winter semester.
- Organized special competitions with similar organizations at other universities.

Soccer Coach (September 2017 – May 2020)

- Conducted weekly practices for a mixture of people with and without intellectual and development disabilities.
- Mentored new members/coaches in how to conduct practices.

- Collaborated with and assisted coaches from other sports (flag football, dance, and basketball) to organize and run weekly practices.

Special Olympics Michigan, Mount Pleasant, MI

April 2016 – June 2019

Bocce Ball Coach

- Resolved conflicts between athletes by implementing creative techniques learned from past experiences.
- Utilized trial-and-error teaching techniques to teach athletic and teamwork skills to athletes.
- Chaperoned the Special Olympics Michigan State Summer Games every June where all participants stayed in dorm halls at Central Michigan University for three days.
- Managed a diverse group of participants that vary in skill gap, age, ethnicity, and economic background.
- Organized practice agendas.

WORK EXPERIENCE

USA Hockey, Mount Pleasant, MI & Ann Arbor, MI

Sept. 2010 – Sept. 2017, Sept. 2018 – March 2020

Referee

- Officiated hockey games with players ranging from elementary students to adults 60 and older.
- Resolved conflicts between coaches and myself by describing my decisions with a stern and professional character.

TEACHING AND MENTORSHIP EXPERIENCE

After-School Family STEM Program at the Goodman South Library, Madison, WI *Sept. 2022 – Nov. 2022*

Teacher

- Co-founded a ten week after-school drop-in family STEM program (K-5 with parents) STEM program in collaboration with the lead youth librarian at the Goodman South Public Library.
- Conducted mini-science experiments in various topics in STEM (e.g., physics, astronomy, biology etc.) with families in an historically underrepresented neighborhood of Madison.
- Created lesson plans adapted for a drop-in style program.
- Increased STEM engagement opportunities to youth and families.

Teaching Assistant at the University of Wisconsin – Madison, Madison, WI

Biology 570 (Cell Biology)

Aug. 2024 – Dec. 2024

- Developed comprehensive lesson plans and facilitated engaging optional weekly discussions.
- Designed and administered weekly Canvas quizzes to reinforce learning outcomes from lectures and reading assignments.
- Provided dedicated support during weekly office hours, addressing student questions and fostering academic growth.
- Graded and provided feedback on >100 papers, ensuring rigorous evaluation and constructive guidance
- Proctored Exams

Biology 151 Lab (Intro Biology Lab Section)

Aug. 2022 – Dec. 2022

- Taught undergraduate students how to conduct scientific research.
- Developed and presented slides to guide students through a three-hour lab section.
- Graded lab reports and provided constructive feedback to develop writing skills of the students.
- Held weekly office hours.
- Proctored exams.

Mentor Undergraduate Researcher, Bement Lab, Madison, WI

Jan. 2023 – Present

- Teach an undergraduate student the background and basics of molecular biology and microscopy.
- Provide feedback on final paper and poster preparation.

Biochemistry (MCDB 310) Study Group Facilitator

- Facilitated a group of students to study weekly in a collaborative environment.
- Utilized learning techniques that encourage members to thoroughly engage in the material.
- Created the group agenda and ensure that group members remain on task.

CMB Graduate Program NSF Workshop Leader, Madison, WI

June 2022 – July 2023

- Utilize knowledge gained from previously applying and winning an honorable mention in the NSF GFRP to mentor graduate students applying for the fellowship.
- Create and present slides with information about the application process and strategies to win the award.
- Review and provide feedback to applicants on their personal and research statements.

PROFESSIONAL TRAINING

Public Engagement with STEM Course (Public Service Fellows), Madison, WI

Jan. 2022 – Dec. 2022

- Participated in a yearlong WIScience course (application required) that focused on how the STEM community can effectively engage with the public.
- Had weekly discussions with a cohort of engaged graduate students from diverse STEM backgrounds.
- Learned about various pathways for individuals in the scientific community to increase public engagement with STEM.
- Learned to view and solve public issues using an asset-based approach instead of a traditional deficit model.
- Discovered the importance of good collaboration between the scientific community and the public to build trust.
- Collaborated with the cohort to determine the best methods and practices to partner with community organizations outside of academia.

WIScience Delta Program Research Mentor Training, Madison, WI

Sept. 2021 – Dec. 2021

- Explored strategies for effective mentoring in a discussion setting.
- Collaborated with peer mentors to problem solve problems with mentor-mentee relationships.

PUBLICATIONS

Chomchai, D., Leda, M., Golding, A., von Dassow, G., Bement, W.M. & Goryachev, A.G. Rho GTPase dynamics distinguish between models of cortical excitability. *Current Biology*, 35(6). doi:10.1016/j.cub.2025.02.003 (2025).

Yang, X., Reist, L. **Chomchai, D.**, Chen, L., Arines, F. & Li, M. ESCRT, not intraluminal fragments, sorts ubiquitinated vacuole membrane proteins for degradation. *Journal of Cell Biology*, 220(8). doi:10.1083/jcb.202012104 (2021).

Yang, X., Zhang, W., Wen, X., Bulinski, P., **Chomchai, D.**, Arines, F., Liu, Y., Sprenger, S., Teis, D., Klionsky, D.J. & Li, M. TORC1 regulates vacuole membrane composition through ubiquitin- and ESCRT-dependent microautophagy. *Journal of Cell Biology*, 219(3). doi:10.1083/jcb.201902127 (2020).

POSTERS

Yang, X., Zhang, W., **Chomchai, D.**, Wen, X., Klionsky, D. & Li M. TORC1 regulates the vacuole membrane proteome through ubiquitin- and ESCRT-dependent microautophagy. **MCDB Department Retreat**, Ypsilanti, MI, Dec. 2019.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **The American Society of Cell Biology – Cell Bio 2022**, Washington D.C., Dec 2022.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **The 19th International Xenopus Conference**, Cambridge, MD, August 2023.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **The American Society of Cell Biology – Cell Bio 2023**, Boston, MA, Dec 2023.

Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **Polarity and Asymmetry in Tissue Architecture and Cell Fate – Gordon Research Conference**, Manchester, NH, June 2024.

Chomchai, D. & Bement, W. Exploring the mechanism of positive feedback in cortical excitability. **The American Society of Cell Biology – Cell Bio 2024**, San Diego, CA, Dec 2024.

Chomchai, D. & Bement, W. Exploring the mechanism of positive feedback in cortical excitability. **The American Society of Cell Biology – Cell Bio 2025**, Philadelphia, PA, Dec 2025.

PRESENTATIONS

Seminar Talk. Chomchai, D. & Bement, W. Exploring the mechanics of positive feedback in Ect2-mediated cortical excitability. **Cells, Science and Imaging (CSI): Madison**. Madison, WI, Oct 2025

Subgroup Talk: Frontiers in Cytokinesis (Ten-minute talk). **Chomchai, D. & Bement, W.** Exploring the mechanism of positive feedback in cortical excitability. **The American Society of Cell Biology – Cell Bio 2024**, San Diego, CA, Dec 2024.

Seminar Talk. Chomchai, D. & Bement, W. Positive feedback between Rho and Ect2 is essential for cortical excitability. **Cell Biology Seminar Group**. Madison, WI, March 2024

Subgroup Talk: Frontiers in Cytokinesis (Ten-minute talk). **Chomchai, D. & Bement, W.** Positive feedback between Rho and Ect2 is essential for cortical excitability. **The American Society of Cell Biology – Cell Bio 2022**, Washington D.C., Dec 2022.

HONORS AND AWARDS

NSF GFRP Honorable Mention, 2022

Bement Lab, University of Wisconsin-Madison, Madison, WI

- Proposal Title: The Basis of Positive Feedback in Cortical Excitability During Cytokinesis

High Honors for Senior Thesis in Molecular, Cellular, and Developmental Biology, 2022

Ming Li Lab, University of Michigan, Ann Arbor, MI

- Thesis Title: The Mechanism of the Ubiquitination and Degradation of Lysosomal Membrane Proteins

Area 7 Outstanding Volunteer of the Year Award, 2017

Special Olympics Michigan, Mount Pleasant, MI

PROFESSIONAL COMMITTEES

Cellular and Molecular Biology Recruitment Committee

Virtual Weekend Recruitment Manager

Jan. 2021 and 2022

- Scheduled a virtual recruitment weekend for the graduate training program
- Organized interviews between recruits and current students
- Created and conducted a virtual social event for recruits and current grad students in Gather.town